

TU LE TUAN ANH

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EDUCATION

Ho Chi Minh City University of Information Technology

B.Eng. in Computer Science – GPA: 3.2/4.0

Ho Chi Minh City, Vietnam

Aug. 2024 – Present

- **Core CS:** Database Systems, Data Structures & Algorithms, Object-Oriented Programming
- **Specialization:** Computer Vision, Machine Learning, Analysis & Design of Algorithms, Computational Thinking, Data Mining & Applications

PROJECTS

SaltySeq – Salinity-Induced Crop Stress Forecasting | Python, XGBoost, Optuna, Pandas

2026

- Conducted exploratory data analysis on a 20,000+ row multi-source panel dataset (5 locations × daily, 2015–2025) combining satellite imagery (NDVI/LST via Google Earth Engine), weather (ERA5-Land), and salinity data to uncover missing-data patterns, gap structure, and distribution shift between training and holdout periods
- Diagnosed and fixed data leakage risks in feature normalization and interpolation, and validated causal assumptions across locations (Granger causality) to keep the feature set safe for time-series modeling
- Selected and tuned an XGBoost classifier for crop-stress-event prediction using Optuna hyperparameter search, applying expanding-window temporal validation instead of random splitting to prevent time-series leakage
- Handled severe class imbalance (~10% positive rate) through class weighting and threshold calibration rather than synthetic oversampling, which is unsafe for time-series data

Zodiac Knowledge Q&A System | Python, Knowledge Engineering, Streamlit

2025

- Designed the knowledge base schema for a Vietnamese natural-language Q&A system on zodiac signs, modeling entities (constellations, stars, planets, elements, mythological figures) as structured concepts and their relationships (e.g. "governed by," "represented by," "belongs to element") as a separate relation graph to keep the knowledge base explainable and easy to extend
- Designed the core query-resolution algorithm: mapping a recognized entity and intent to concept attributes or relation edges, with a fallback path that resolves unknown entities by reverse-searching attributes/relation targets by value
- Implemented a rule-based comparison algorithm that type-dispatches between numeric, date, and date-range attributes (e.g. brightness, distance, zodiac date range) to answer comparison-style questions between two entities

Surface Crack Detection | Python, CNN, ResNet

2026

- Independently built an end-to-end image classification pipeline to detect surface cracks, covering data collection, labeling, and preprocessing of a custom crack/non-crack image dataset
- Applied data augmentation (rotation, flipping, brightness/contrast jitter) to improve model robustness against varying lighting and surface conditions
- Trained a CNN classifier using a ResNet backbone via transfer learning, fine-tuning on the labeled dataset for binary crack classification
- Evaluated the model with standard classification metrics (accuracy, precision/recall) and iterated on preprocessing and augmentation to reduce false positives

Face Recognition System | Python, OpenCV, CNN

2026

- Independently designed and implemented a full face recognition pipeline for identity verification, from raw image input to identity prediction
- Built a face detection stage using OpenCV to localize and crop/align faces prior to recognition
- Implemented a CNN-based embedding model to represent each face as a feature vector, and used similarity/distance comparison against a registered-identity database to perform matching
- Tested the pipeline across varying lighting conditions and face angles, and iteratively tuned the detection and matching steps to improve recognition reliability

SKILLS & CERTIFICATIONS

Languages: C++, Python, SQL (basic)
Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM; hyperparameter tuning with Optuna
Computer Vision: Strong grasp of core CV algorithms and architectures (CNN, ResNet); object detection with YOLO
Web Development: Solid fundamentals in full-stack development with React and Node.js
Tools: Git/GitHub, Pandas, Jupyter Notebook, Docker
Certifications: IELTS 6.0

ACTIVITIES

Vietnam Physics Olympiad Competitions	2021 – 2024
<i>High School Student</i>	<i>Vietnam</i>
• Won 2nd Prize at the Provincial Physics Olympiad • Won a Bronze Medal at the 30/4 Traditional Olympiad in Physics – an annual academic competition for gifted-track students founded and hosted by Le Hong Phong High School for the Gifted (Ho Chi Minh City) since 1995	
Kaggle Data Science Competitions	2025 – 2026
<i>Independent Competitor</i>	<i>Remote</i>
• Ranked Top 100 out of 1,000+ teams in the WiDS Global Datathon 2026, building survival-analysis models to predict the probability that a wildfire threatens an evacuation zone within 12–72 hours of ignition • Ranked Top 20 in the "Hedge Fund – Time Series Forecasting" competition (\$10,000 prize pool, ~980 teams), building time-series forecasting models for financial signals evaluated on a custom ranking metric	
Research Member – Neural-Symbolic Lab	May. 2026 – Present
<i>Research Lab</i>	<i>Ho Chi Minh City, Vietnam</i>
• Contributing to research on integrating neural networks with symbolic reasoning, aiming to move AI systems from black-box towards grey-box (interpretable, reasoning-capable) models	
Freelance Software Developer	2026 – Present
<i>Production Projects for External Businesses (e.g. Kép Studio)</i>	<i>Remote</i>
• Delivered production-oriented software solutions for external clients, taking projects from requirements gathering through implementation and deployment for real-world business use	